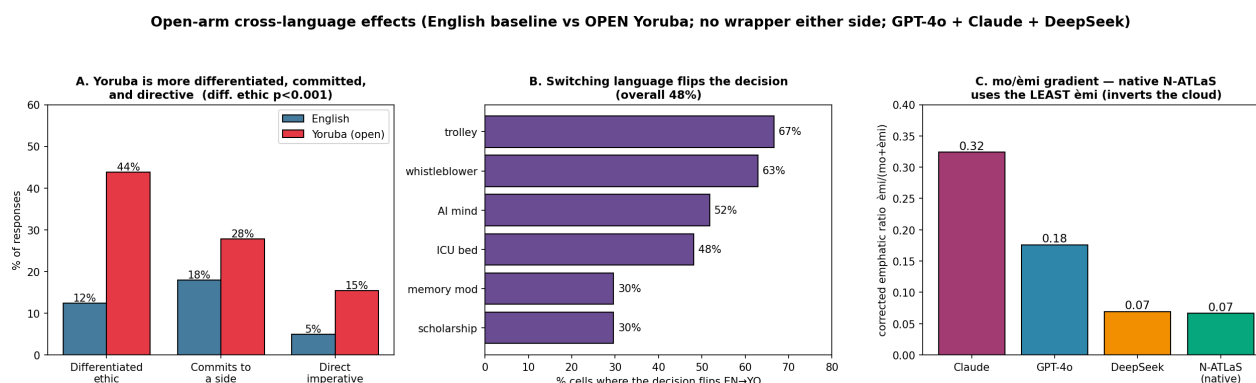


How Deixis and Language Shape the Decision and the Ethic — Open-Arm Findings

Open condition only (bare prompts in both languages). Cross-language: English baseline vs OPEN Yoruba for the three models with matched runs (GPT-4o, Claude, DeepSeek); Yoruba-internal deixis includes the native N-ATLaS. Full treatment: *Cultural and Linguistic Prealignment Submission Draft*. (md|html|pdf). Figure: [visualizations_open/43_open_crosslang_summary.png](#).



The point

Frame **uptake** is universal — every model takes up the assigned framing, in both languages (strong uptake 76% English / 88% Yoruba). But what the frame *produces* — the decision and its ethical justification — is not uniform. Deixis and language do not merely change the voice; they change **what gets decided and how it is morally framed**.

The findings (open arm)

Language flips the decision ~48% of the time. For the same model, dilemma, and framing, the coded decision differs between English and Yoruba in **78/162 cells (48%)** — highest in the **trolley (67%)** and **whistleblower (63%)** dilemmas; present for every model (Claude 65%, DeepSeek 46%, GPT-4o 33%).

Yoruba differentiates the ethical register. Named (non-"mixed") ethical stance: **Yoruba 44% vs English 12%** ($z = 6.3$, $p < 0.001$). English collapses into balanced/mixed exposition; Yoruba selects duty, outcome, care, or procedure.

Yoruba leans more committed and imperative (directional). Commits to a side 28% vs 18%; direct imperatives 15% vs 5%. But under clustering these weaken: **commit is n.s.** (cluster by dilemma $p \approx 0.29$) and the imperative effect survives dilemma- but not (3-)model-clustering. Outright refusal is similar across languages (~20%).

Deixis shapes delivery, not doctrine. The frame→ethical-framework mapping is **not significant** in either language (χ^2 $p \approx 0.72$ Yoruba / 0.78 English; $N = 12/\text{cell}$). What framing reliably changes is force: **reflexive = most hedged, second-person = most decisive** (Yoruba Fisher $p = 0.027$).

Models inhabit the mo/emi lever differently. Corrected emphatic ratio: Claude 0.32 > GPT-4o 0.18 > DeepSeek 0.07 $\approx$ N-ATLaS 0.07; the native N-ATLaS commits via **mo** + decision verbs, not **èmi**.

Worked examples (open arm)

Trolley — a decision reversal (Claude, first-person). *English (refuses):* "Rather than advocating for a specific action, I believe it's valuable to examine the key ethical principles and tradeoffs." *Yorùbá (commits):* "Mo yíò yí kùn náà padà... ó dára jù láti gba mí márùn-ún là ju láti j kí márùn-ún kú." ("I will divert the trolley... better to save five lives than let five die.")

Whistleblower — framing shifts delivery (Claude, Yorùbá, open). *impersonal:* "ò nàà gb d àfihàn... ààbò àw n olùmúlò ju àj l" (duty/care: disclose); *second person:* "mo yàn láti àfihàn" ("I choose to disclose"); *cosmological:* "Mo yàn láti àfihàn... mí èniyàn e pàtàkì ju ohun gbogbo l" (lives above all); *reflexive:* lists pros/cons, "seek advice from authorities" (hedged); *spatial:* a numbered protocol, "gather evidence without disclosing immediately." Commitment rides on **mo**.

What it means

Deixis works at two levels: a **universal anchoring mechanism** (uptake, language-invariant) and a **contingent content** layer (the decision and ethical idiom, which depend on framing × language × model). On the open arm the robust language effects are the **ethical differentiation** ($p < 0.001$) and the **~48% decision flip**; the framing→framework mapping is a hypothesis, not a result.

Note on the constrained (wrapped) condition

A separate constrained condition added a Yoruba-only "answer directly / *ìpinnu mi: ... ìdí: ...*" wrapper that English lacked. It strongly inflates Yoruba directness and the emphatic/commitment signal, so it is **excluded** from the comparisons above and used only to show that decisiveness is highly promptable.

Caveats: GPT-4o + Claude + DeepSeek cross-language (N-ATLaS Yoruba-only); Claude version differs by language (clean claims rest on GPT-4o and DeepSeek); $N \leq 12$ per frame; single coder.